Synthesis L. Tual et al. Paper

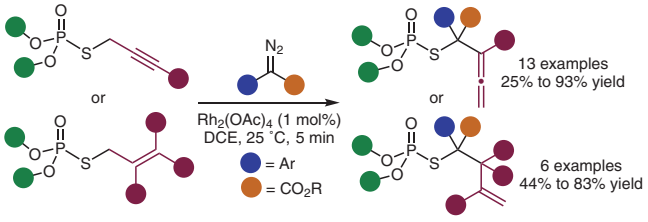
Doyle–Kirmse Reaction of *S*-Allyl and *S*-Propargyl Phosphorothioates

Laurine Tual◊
Charlou Rognan◊
Timothé Maujean◊
Romain Pertschi
Nicolas Girard
Gaëlle Blond
Mihaela Gulea*

Université de Strasbourg, CNRS, Laboratoire d'Innovation Thérapeutique, LIT UMR 7200, 67000 Strasbourg, France
gulea@unistra.fr

◊ These authors contributed equally

Published as part of the *Special Topic C–S Bond Forming and Cleaving Reactions*



Received: 22.11.2024
Accepted after revision: 11.01.2025
Published online: 11.01.2025 (Accepted Manuscript), 25.02.2025 (Version of Record)
DOI: 10.1055/a-2515-0351; Art ID: SS-2024-11-0461-OP

Abstract The Doyle–Kirmse reaction represents a powerful synthetic tool for forming both C–S and C–C bonds in a single process. The rhodium-catalyzed Doyle–Kirmse reaction of *S*-allyl and *S*-propargyl phosphorothioates with diazoesters has been developed and represents a rare example involving substrates bearing an electron-withdrawing group on the sulfur atom. The transformation takes place at room temperature, in only 5 minutes, and leads via a [2,3]-sigmatropic rearrangement of the sulfonium ylide intermediate to the corresponding *S*-homoallyl or *S*-homoallenyl phosphorothioates in yields ranging from 25% to 92% (19 examples).

Key words Doyle–Kirmse reaction, [2,3]-rearrangement, propargyl sulfides, allyl sulfides, diazoesters, phosphorothioates, C–S bond formation

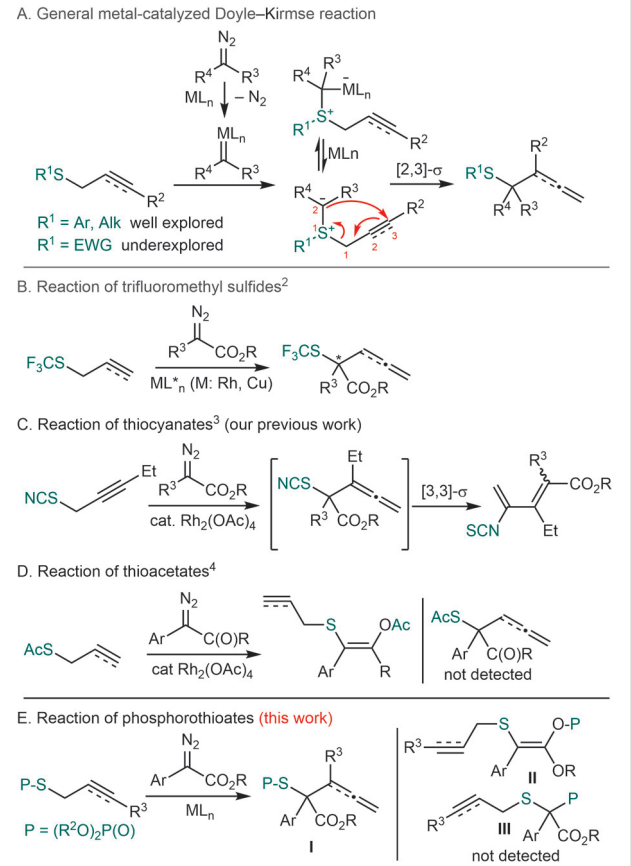
The Doyle–Kirmse reaction is a powerful method for forming both C–S and C–C bonds in a single process, enabling the synthesis of a wide range of homoallyl or homoallenyl sulfides.[1] The metal-catalyzed version consists of the reaction of an allyl or a propargyl sulfide and a metal carbene species classically generated by the decomposition of a diazo compound (Scheme 1, A). This leads to the key intermediate of this reaction which is a sulfonium ylide that can be either metal-bound or metal-free depending mainly on the nature of the substrate and metal. The resulting ylide can then evolve through a [2,3]-sigmatropic rearrangement to the corresponding homoallyl or homoallenyl sulfide. Various transition-metal complexes, mainly based on rhodium and copper, but also on ruthenium, iron, cobalt, silver, platinum and gold, have been used in this reaction. The use of chiral metal complexes made possible enantioselective versions of the reaction.[1g] More recent research described a metal-free Doyle–Kirmse reaction using light-mediated diazo decomposition that generates free carbene species.[1h,i] A literature survey revealed that among the different substrates used in this reaction, aryl and alkyl sulfides represent the overwhelming majority, while derivatives bearing an electron-withdrawing group (EWG) on the sulfur have been sparsely reported (Scheme 1, A). This may be due to the concern that the sulfur, depleted by an electron-withdrawing substituent, would be less reactive toward the electrophilic metal carbene species. However, allyl and propargyl trifluoromethyl sulfides were successfully used in enantioselective Doyle–Kirmse reactions employing chiral rhodium or copper complexes (Scheme 1, B).[2] Another example, that we recently reported, concerned propargyl thiocyanates (Scheme 1, C).[3] In this case, the products were not isolated, as they subsequently rearranged to the isothiocyanate derivatives via a thio-Claisen reaction. Interestingly, when *S*-allyl or *S*-propargyl thioacetates were reacted with aryl diazoketones, the *S*-ylide formation was followed by sulfur-to-oxygen acyl transfer, and not by the [2,3]-rearrangement, thus leading to alkenyl sulfides instead of the expected products (Scheme 1, D).[4]

These literature data prompted us to investigate the reactivity of *S*-allyl and *S*-propargyl phosphorothioates, as new electron-poor sulfide substrates, toward α-diazocarbonyl compounds.

Moreover, the phosphorothioates, which belong to the class of P(V) organophosphorus compounds,[5] have found important applications as biologically active molecules in agrochemical and pharmaceutical fields.[6] Their preparation, as well as their uses as synthetic intermediates in various transformations, most of which involving the cleavage of the P–S or C–S bond, have been independently reviewed in 2018 by O'Sullivan and co-workers, and by our group.[7] Two other reviews that followed dealt with modern methods to access phosphorothioates by phosphorus–sulfur bond formation and by direct phosphorothiolation reactions, respectively.[8] A recently reported synthesis of phos-

Downloaded by: Fudan University Library. Copyrighted material.

© 2025. Thieme. All rights reserved.



Scheme 1 Reported Doyle–Kirmse reactions with various sulfide derivatives and our study of phosphorothioates

phorodithioates used a catalyst- and light-free reaction of phosphorodithioic acids with diazo compounds.[9]

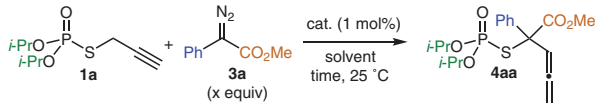
The reaction of a phosphorothioate with the metallocarbene resulting from diazo compound decomposition would, in the first step, lead to a phosphoryl sulfonium ylide which, to our knowledge, is still an unknown species. The so-formed *S*-ylide can then evolve along three possible pathways: [2,3]-sigmatropic rearrangement yielding the Doyle–Kirmse reaction product **I**, S-to-O phosphoryl transfer (similar to the acyl transfer), leading to the alkenyl phosphate **II**, or S-to-C phosphoryl group migration,[10] resulting in the α-sulfanyl phosphonate **III** (Scheme 1, E). Herein, we describe the rhodium-catalyzed reaction of *S*-allyl and *S*-propargyl phosphorothioates with diazoesters, leading exclusively to the Doyle–Kirmse [2,3]-rearrangement products **I**.

To set up the reaction conditions, easily available *O*,*O*-diisopropyl *S*-propargyl phosphorothioate (**1a**) was selected as a model substrate. Initial experiments were performed by adding 2-diazo-2-phenylacetate **3a** (1.5 equiv) to a solution of **1a** (1.0 equiv) in toluene (c = 0.1 M), in the presence of $Rh_2(OAc)_4$ (1 mol%) as a catalyst, to ensure effective diazo decomposition to generate rhodium carbene (Table 1, entry 1). After only 10 minutes, ^{1}H NMR spectroscopic analysis of the crude mixture revealed full conversion of the diazo partner **3a** and the expected homoallenyl phosphorothioate **4aa** was identified as the major product of a complex crude reaction mixture (31%, ^{1}H NMR yield) and was isolated in 31% yield. With this result in hand, optimization studies were conducted further. First, the nature of the ligand on the rhodium dimer catalyst was investigated, switching from acetate (OAc) to the bulky pivalate (OPiv) or to the electro-deficient trifluoroacetate (TFA). The use of the first complex led to a higher conversion compared to the acetate, however a lower ^{1}H NMR yield was measured (entry 2 *vs* 1), while the second complex led to a lower conversion and yield (entry 3 *vs* 1). We selected the $Rh_2(OAc)_4$ for further experiments, as it represented the best compromise between conversion and yield. Then, the influence of the number of equivalents of diazoester partner **3a** was investigated in an attempt to increase the conversion of **1a** to completion. Indeed, increasing from 1.5 to 2.0 equivalents of **3a** substantially improved conversion and yield (entry 1 *vs* 4 and 5). With 4.0 equivalents full conversion (>99%) of the starting phosphorothioate **1a** was reached and **4aa** was obtained in 70% yield (entry 6). To further improve the yield of the transformation, the impact of the solvent was investigated. To our surprise, a switch from toluene to dichloromethane was detrimental to the reaction as only 12% of **4aa** was actually obtained and a fast decomposition and dimerization of the diazo partner was observed (entry 7). Dichloroethane (DCE) was then tested and resulted in a smooth diazo decomposition with full conversion (>99%) of the starting material **1a**, leading to **4aa** in a good isolated yield of 85% (entry 8). It is worth noting that the use of DCE also reduced the reaction time from 10 minutes to 5 minutes. When lowering the catalytic charge to only 0.1 mol%, the reaction was still efficient, with a conversion of 98% and an NMR yield of 80% (entry 9). The nature of the metal catalyst was also investigated by testing the ruthenium(II) complex $[(\eta^6\text{-}C_6H_6)RuCl(\mu\text{-}Cl)]_2$, copper thiophene-2-carboxylate, silver triflate and palladium(II) chloride, but in all cases no conversion of the starting material **1a** was observed (Table 1, entries 10–13). In the case of the Ru, Ag and Pd catalysts, diazoester decomposition occurred and led to dimerization of the carbene, while with CuTC the diazoester was recovered. Finally, we kept the reaction conditions given in entry 8 for our further experiments.

With these optimal conditions in hand, several structural modifications of the reaction partners were investigated to explore the scope and limitations of the transformation (Scheme 2). Changing the isopropyl groups on the phosphoryl of **1** to ethyl or L-menthyl groups did not change the reactivity, leading to similar yields. Unfortunately, the asymmetric Doyle–Kirmse reaction using dimenthyl phosphorothioate **1c** as the chiral substrate was not diaste-

Downloaded by: Fudan University Library. Copyrighted material.

Table 1 Optimization of the Doyle–Kirmse Reaction of Phosphorothioate **1a** with Diazoester **3a**



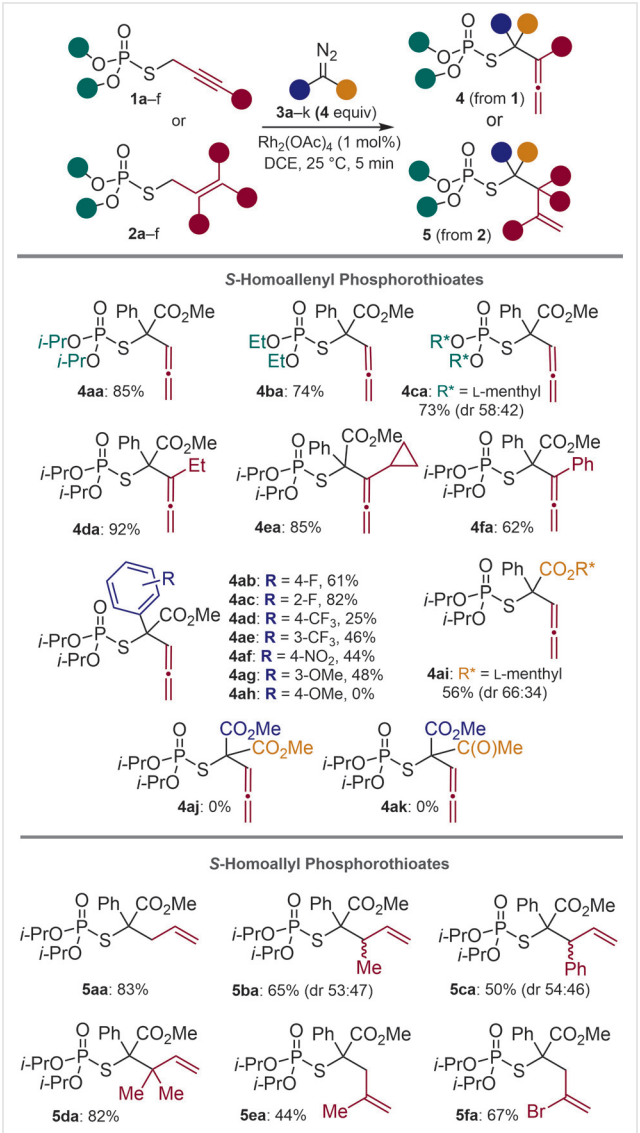
Entry	**3a** (equiv)	Catalyst	Solvent	Time	Conversion (%)[a]	Yield (%)[b]
1	1.5	$Rh_2(OAc)_4$	PhMe	10 min	36	31 (31)[c]
2	1.5	$Rh_2(OPiv)_4$	PhMe	10 min	53	21
3	1.5	$Rh_2(TFA)_4$	PhMe	40 min	29	7
4	2	$Rh_2(OAc)_4$	PhMe	10 min	52	47
5	3	$Rh_2(OAc)_4$	PhMe	10 min	89	70
6	4	$Rh_2(OAc)_4$	PhMe	10 min	>99	70
7	4	$Rh_2(OAc)_4$	CH_2Cl_2	5 min	13	12
8	**4**	**$Rh_2(OAc)_4$**	**DCE**	**5 min**	**>99**	**82 (85)[c]**
9[d]	4	$Rh_2(OAc)_4$	DCE	5 min	98	80
10[e,i]	1.5	Ru(II)	PhMe	24 h	0	0
11[f,j]	1.5	CuTC	PhMe	24 h	0	0
12[g,i]	1.5	AgOTf	PhMe	24 h	0	0
13[h,i]	1.5	$PdCl_2$	PhMe	24 h	0	0

[a] Conversion relative to **1a** determined by ^{1}H NMR with caffeine as internal standard.
[b] Yield of **4aa** determined by ^{1}H NMR with caffeine as internal standard.
[c] Isolated yield of **4aa** in brackets.
[d] 0.1 mol% of catalyst was used.
[e] 1 mol% of $[(\eta^6\text{-}C_6H_6)RuCl(\mu\text{-}Cl)]_2$ catalyst was used.
[f] 10 mol% of catalyst was used.
[g] 10 mol% of catalyst was used.
[h] 5 mol% of catalyst was used.
[i] Decomposition and dimerization of the diazoester observed.
[j] Diazoester recovered.

reoselective, leading to a very low diastereomeric ratio of 58:42. The chiral substrate **1c** was also reacted using a chiral Rh(II) catalyst derived from (*S*)-*N*-naphthalene-2,3-dicarboxyl-*tert*-leucine [Rh_2(*S*-BPTTL)$_4$],[11] but the diastereomeric ratio and the NMR yield of the product **4ca** remained the same. This indicates that no chiral information was transferred from the metal complex to the substrate, suggesting that the rearrangement occurs via an ylide-free pathway.

Next, the structure of the propargyl moiety was modified. We prepared substrates **1d–f** bearing an ethyl, a cyclopropyl or a phenyl substituent, respectively, on the triple bond. They were reacted with diazoester **3a** under the standard reaction conditions, and all of them were successfully transformed into the desired products **4**. Ethyl and cyclopropyl derivatives **4da** and **4ea** were isolated in very good yields of 92% and 85%, respectively, whereas phenyl derivative **4fa** was obtained in 62% yield.

Modification of the substituent on the aryl group of the diazo derivative was then considered. Using methyl 2-aryl-2-diazoacetates **3b–g** in reactions with phosphorothioate **1a**, the desired homoallenyl phosphorothioates **4** were obtained, regardless of the position or electronic nature of the substituent on the aromatic ring. The only exception was the 4-OMe derivative **3h**, which decomposed and dimerized, leaving substrate **1a** intact. The use of diazoesters **3d–f** with an electron-withdrawing group such as CF_3 and NO_2 reduced the efficiency of the reaction leading to lower yields (25% to 46%), which may be attributed to the increased electrophilicity and reactivity of the corresponding metal carbenes. The L-menthyl diazoacetate **3i** allowed us to obtain product **4ai** in 56% yield and a diastereoselectivity



Scheme 2 Scope and limitations of the Doyle–Kirmse reaction of phosphorothioates **1** and **2** with diazoesters **3**

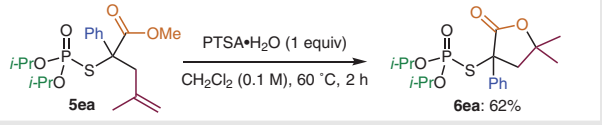
Downloaded by: Fudan University Library. Copyrighted material.

of 66:34. Acceptor–acceptor diazo derivatives such as dimethyl diazomalonate or acetyl diazoacetate proved to be ineffective, as the expected homoallenyl phosphorothioates **4aj** and **4ak** were not observed.

We then examined the efficiency of the reaction in the *S*-allyl series, by using different phosphorothioates **2a–f** and the diazoester partner **3a**. Starting from substrate **2a**, the *S*-homoallyl derivative **5aa** was obtained in a good yield of 83%. When *S*-crotyl **2b** (77:23 ratio of *E*/*Z*-isomers) and *S*-cinnamyl phosphorothioates **2c** (100% *E*-isomer) were used, slightly lower yields were reached, 65% for **5ba** and 50% for **5ca**. In both cases, no diastereoselectivity was observed, and products **5ba** and **5ca** were obtained as mixtures of diastereomers in ~1:1 ratio. On the other hand, *S*-dimethylhomoallyl phosphorothioate **5da** was obtained in a good yield of 82% from the *S*-prenyl precursor **2d**, without any detection of deprenylated byproducts in the crude reaction mixture, as previously reported for the phenyl sulfide series.[12] The *S*-methallyl **2e** and *S*-bromoallyl **2f** substrates also led to the corresponding *S*-homoallyl phosphorothioates, in 44% and 67% yield, respectively.

To showcase the robustness and practicality of our method, the reaction was scaled up to 1 mmol of sulfide **1a** and 4 mmol of diazoester **3a**, using only 0.1 mol% of Rh catalyst. In this case, 20 minutes was necessary for a complete conversion and phosphorothioate **4aa** was isolated in 64% yield.

Taking advantage of the presence of the carboxylate and the double bond functionalities in the obtained products, we engaged an *S*-homoallyl phosphorothioate in an intramolecular Brønsted acid catalyzed hydroacyloxylation to access a lactone. By treating phosphorothioate **5ea** with 1 equivalent of *p*-toluenesulfonic acid (PTSA) monohydrate, 5-*exo*-trig cyclization was achieved efficiently, leading to the expected lactone **6ea** in 62% yield (Scheme 3).



Scheme 3 Intramolecular hydroacyloxylation to access lactone **6ea**

In summary, we have developed a rhodium-catalyzed Doyle–Kirmse reaction of *S*-allyl and *S*-propargyl phosphorothioates with aryl diazoacetates. This represents a rare example of Doyle–Kirmse reaction involving derivatives bearing an electron-withdrawing group on sulfur. The transformation occurs under mild conditions, at room temperature, in only 5 minutes, and exhibits a good substrate tolerance. It led via the [2,3]-sigmatropic rearrangement of sulfonium ylides to various *S*-homoallyl and *S*-homoallenyl phosphorothioates (19 examples) in yields ranging from 25% to 92%. To illustrate the potential post-functionalization of the obtained products, a phosphorothioate-substituted lactone was prepared by acid-catalyzed cyclization of the carboxylate function onto the homoallyl double bond. From an organic synthesis point of view, this reaction is a practical method toward accessing a series of compounds containing a quaternary carbon stereocenter bearing a phosphorothioate functional group, which is of interest in drug design, and as a route to introduce P and/or S into molecules.

All reagents, chemicals and anhydrous solvents were purchased from commercial sources and used without purification. Reactions were monitored by TLC using Merck silica gel 60 F254 on aluminum sheets. TLC plates were visualized under UV light and revealed with acidic *p*-anisaldehyde stain or vanillin stain. Products were purified by flash column chromatography on Merck silica gel 60 (40–63 μm) for normal phase or C18 for reverse phase. NMR spectra were recorded in $CDCl_3$ on a Bruker spectrometer at 400 MHz for ^1H, at 101 MHz for ^{13}C, at 162 MHz for ^{31}P and at 376 MHz for ^{19}F. Proton-decoupled fluorine and phosphorus NMR spectra, $^{19}F\{^1H\}$ NMR and $^{31}P\{^1H\}$ NMR, were performed for all fluorinated and phosphorylated compounds. Proton chemical shifts are reported in ppm (δ), relative to residual $CHCl_3$ (δ 7.26 ppm). Multiplicities are reported as follows: singlet (s), doublet (d), triplet (t), quartet (q), heptet (hept), multiplet (m) and combinations thereof. Coupling constant values *J* are given in Hz. Carbon chemical shifts are reported in ppm (δ), relative to the internal standard $CDCl_3$ (δ 77.23 ppm). ^1H and ^{13}C NMR signals were assigned mostly on the basis of 2D-NMR (COSY, HSQC, HMBC) experiments. Diastereomeric ratios were determined according to ^1H NMR of the crude and NMR yields were determined using caffeine as internal standard in ^1H NMR. High-resolution mass spectral analyses were performed using an Agilent 1200 RRLC HPLC chain and an Agilent 6520 Accurate-Mass Q-ToF system.

Compounds 4 and 5; General Procedure

To a solution of substrate **1** or **2** in DCE (0.05 M) were added $Rh_2(OAc)_4$ (1 mol%) and a solution of diazoester **3** (4 equiv) in DCE (0.05 M). The reaction mixture was then stirred at room temperature until completion, and the solvent was removed under vacuum. The crude was purified by flash chromatography on silica gel (*n*-pentane/Et_2O) to get pure product **4** or **5**. A second purification using reverse-phase chromatography was performed for some examples difficult to purify on silica gel. The residue was dissolved in DMSO (0.5 mL), injected on a C18 flash column and purified using a gradient of MeCN in H_2O (30% to 100% MeCN in 30 min).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-phenylpenta-3,4-dienoate (4aa)

According to the general procedure, starting from **1a** and **3a**, product **4aa** (69 mg, 0.18 mmol, 85%) was obtained as a yellow oil.

R_f = 0.16 (*n*-pentane/Et_2O, 7:3).

^1H NMR (400 MHz, $CDCl_3$): δ = 7.53–7.47 (m, 2 H, H^6), 7.36–7.25 (m, 3 H, H^7 + H^8), 6.37 (t, *J* = 6.6 Hz, 1 H, H^3), 4.83 (dd, *J* = 6.6, 2.6 Hz, 2 H, H^4), 4.75 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 4.60 (dhept, *J* = 8.8, 6.2 Hz, 1 H, H^2), 3.80 (s, 3 H, H^5), 1.34 (d, *J* = 6.2 Hz, 3 H, H^1), 1.29 (d, *J* = 6.2 Hz, 3 H, H^1), 1.28 (d, *J* = 5.9 Hz, 3 H, H^1), 1.24 (d, *J* = 6.2 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 208.0 (C^9), 171.0 (d, *J* = 5.4 Hz, C^{11}), 138.6 (d, *J* = 7.6 Hz, C^{12}), 128.1 (2 C, C^6 or C^7), 128.1 (C^8), 127.8 (2 C, C^6 or C^7), 96.0 (d, *J* = 2.2 Hz, C^3), 79.3 (C^4), 72.9 (d, *J* = 6.4 Hz, C^2), 72.8 (d, *J* = 7.2 Hz, C^2), 63.7 (d, *J* = 3.3 Hz, C^{10}), 53.4 (C^5), 23.9 (d, *J* = 3.7 Hz, C^1), 23.7 (d, *J* = 4.2 Hz, C^1), 23.6 (d, *J* = 6.2 Hz, C^1), 23.5 (d, *J* = 6.3 Hz, C^1).

Downloaded by: Fudan University Library. Copyrighted material.

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.71.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{26}O_5PS$: 385.1239; found: 385.1253 (Δ_{HRMS} = 4.64 ppm).

Methyl 2-((Diethoxyphosphoryl)thio)-2-phenylpenta-3,4-dienoate (4ba)

According to the general procedure, starting from **1b** and **3a**, product **4ba** (58 mg, 0.16 mmol, 74%) was obtained as a yellow oil.

R_f = 0.12 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.42 (dt, *J* = 8.0, 1.2 Hz, 2 H, H^6), 7.31–7.16 (m, 3 H, H^7 + H^8), 6.28 (t, *J* = 6.6 Hz, 1 H, H^3), 4.79 (dd, *J* = 6.6, 1.3 Hz, 2 H, H^4), 4.15–3.80 (m, 4 H, H^2), 3.71 (d, *J* = 1.3 Hz, 3 H, H^5), 1.24–1.13 (m, 6 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 207.9 (C^9), 170.9 (d, *J* = 5.9 Hz, C^{11}), 138.7 (d, *J* = 7.2 Hz, C^{12}), 128.2 (C^8), 128.2 (2 C, C^6 or C^7), 127.7 (2 C, C^6 or C^7), 95.5 (d, *J* = 2.1 Hz, C^3), 79.4 (C^4), 63.9 (d, *J* = 6.0 Hz, C^{10}), 63.9 (d, *J* = 6.0 Hz, C^2), 63.7 (d, *J* = 6.8 Hz, C^2), 53.5 (C^5), 16.0 (d, *J* = 5.7 Hz, C^1), 15.9 (d, *J* = 6.0 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 21.63.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{16}H_{22}O_5PS$: 357.0926; found: 357.0935 (Δ_{HRMS} = 3.95 ppm).

Methyl 2-((Bis(((1*S*,2*R*,5*S*)-2-isopropyl-5-methylcyclohexyl)oxy)-phosphoryl)thio)-2-phenylpenta-3,4-dienoate (4ca)

According to the general procedure, starting from **1c** and **3a**, product **4ca** (98 mg, 0.18 mmol, 73%) was obtained as a yellow oil after silica gel chromatography (0–30% EtOAc in *n*-heptane) as a mixture of non-separable diastereoisomers.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.53–7.44 (m, 2 H, 2 H^{13}), 7.36–7.23 (m, 3 H, 2 H^{14} + 1 H^{15}), 6.41 (t, *J* = 6.6 Hz, H^{10}-*minor*), 6.39 (t, *J* = 6.6 Hz, H^{10}-*major*), 4.86–4.80 (m, 2 H^{11}-*minor*), 4.79–4.72 (m, 2 H^{11}-*major*), 4.30–4.07 (m, 2 H^1), 3.80 (s, 3 H^{12}-*major*), 3.77 (s, 3 H^{12}-*minor*), 2.42–2.01 (m, 4 H), 1.69–1.59 (m, 4 H), 1.48–1.22 (m, 4 H), 1.18–0.94 (m, 4 H), 0.9–0.86 (m, 11 H), 0.86–0.70 (m, 8 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 208.1 (C^{16}-*major*), 208.0 (C^{16}-*minor*), 171.3 (d, *J* = 4.4 Hz, C^{19}-*major*), 171.1 (d, *J* = 5.6 Hz, C^{19}-*minor*), 138.8 (d, *J* = 8.1 Hz, C^{17}-*minor*), 138.4 (d, *J* = 8.7 Hz, C^{17}-*major*), 128.1 (C^{15}), 128.1 (2 C, C^{14}), 127.9 (d, *J* = 1.0 Hz, C^{13}-*minor*), 127.8 (d, *J* = 1.1 Hz, C^{13}-*major*), 96.9 (d, *J* = 2.3 Hz, C^{10}-*major*), 96.3 (d, *J* = 1.9 Hz, C^{10}-*minor*), 79.5–79.1 (m, 4 C, C^{11} + C^1), 64.0 (d, *J* = 3.1 Hz, C^{18}-*minor*), 63.8 (d, *J* = 3.4 Hz, C^{18}-*major*), 53.4 (C^{12}-*major*), 53.3 (C^{12}-*minor*), 48.8–48.4 (m, 2 C), 42.9 (1 C-*minor*), 42.7 (1 C-*major*), 42.6 (d, *J* = 1.2 Hz, 1 C-*minor*), 42.5 (d, *J* = 1.6 Hz, 1 C-*major*), 34.2–34.0 (m, 2 C), 31.7, 31.6, 25.4 (1 C-*major*), 25.3 (1 C-*minor*), 25.3 (1 C-*minor*), 25.2 (1 C-*major*), 23.0–22.8 (m, 2 C), 22.0–21.9 (m, 2 C), 21.2, 21.1 (1 C-*major*), 21.0 (1 C-*minor*), 16.1 (1 C-*minor*), 15.9 (1 C-*major*), 15.87 (1 C-*minor*), 15.7 (1 C-*major*).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.12 (*minor*), 19.47 (*major*).

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{32}H_{50}O_5PS$: 577.3117; found: 577.3145 (Δ_{HRMS} = 4.85 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-3-ethyl-2-phenylpenta-3,4-dienoate (4da)

According to the general procedure, starting from **1d** and **3a**, product **4da** (73 mg, 0.18 mmol, 92%) was obtained as a yellow oil.

R_f = 0.20 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.71–7.65 (m, 2 H, H^7), 7.34–7.20 (m, 3 H, H^8 + H^9), 5.12 (t, *J* = 3.8 Hz, 2 H, H^5), 4.74 (dhept, *J* = 8.9, 6.2 Hz, 1 H, H^2), 4.24 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 3.82 (s, 3 H, H^6), 1.99–1.79 (m, 2 H, H^3), 1.32 (d, *J* = 6.2 Hz, 3 H, H^1), 1.27 (d, *J* = 6.2 Hz, 3 H, H^1), 1.10 (d, *J* = 6.3 Hz, 3 H, H^1), 1.08 (d, *J* = 6.3 Hz, 3 H, H^1), 0.97 (t, *J* = 7.3 Hz, 3 H, H^4).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 206.7 (C^{10}), 170.1 (d, *J* = 3.2 Hz, C^{13}), 139.5 (d, *J* = 3.5 Hz, C^{14}), 129.0 (2 C, C^7 or C^8), 127.7 (2 C, C^7 or C^8), 127.5 (C^9), 108.0 (d, *J* = 9.8 Hz, C^{11}), 81.8 (C^5), 72.6 (d, *J* = 6.5 Hz, C^2), 72.3 (d, *J* = 6.9 Hz, C^2), 65.4 (C^{12}), 53.0 (C^6), 23.8 (d, *J* = 3.6 Hz, C^1), 23.6 (d, *J* = 2.6 Hz, C^1), 23.5 (d, *J* = 3.2 Hz, C^1), 23.4 (d, *J* = 3.0 Hz, C^1), 21.8 (C^3), 12.3 (C^4).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.08.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{20}H_{30}O_5PS$: 413.1552; found: 413.1562 (Δ_{HRMS} = 3.58 ppm).

Methyl 3-Cyclopropyl-2-((diisopropoxyphosphoryl)thio)-2-phenylpenta-3,4-dienoate (4ea)

According to the general procedure, starting from **1e** and **3a**, product **4ea** (63 mg, 0.15 mmol, 85%) was obtained as a pale-yellow oil.

R_f = 0.16 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.82–7.76 (m, 2 H, H^7), 7.37–7.21 (m, 3 H, H^8 + H^9), 5.12 (d, *J* = 1.8 Hz, 2 H, H^5), 4.74 (dhept, *J* = 9.0, 6.2 Hz, 1 H, H^2), 4.19 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 3.84 (s, 3 H, H^6), 1.32 (d, *J* = 6.2 Hz, 3 H, H^1), 1.26 (d, *J* = 6.2 Hz, 3 H, H^1), 1.00 (d, *J* = 6.3 Hz, 3 H, H^1), 1.00 (d, *J* = 6.1 Hz, 3 H, H^1), 0.99–0.95 (m, 1 H, H^3), 0.69–0.57 (m, 2 H, H^4), 0.55–0.48 (m, 1 H, H^4), 0.38–0.31 (m, 1 H, H^4).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 204.6 (C^{10}), 170.1 (d, *J* = 3.1 Hz, C^{13}), 139.7 (d, *J* = 2.9 Hz, C^{14}), 129.4 (2 C, C^7 or C^8), 127.6 (2 C, C^7 or C^8), 127.5 (C^9), 110.1 (d, *J* = 11.0 Hz, C^{11}), 82.6 (C^5), 72.5 (d, *J* = 6.3 Hz, C^2), 72.3 (d, *J* = 6.8 Hz, C^2), 65.1 (d, *J* = 3.2 Hz, C^{12}), 53.0 (C^6), 23.8 (d, *J* = 3.5 Hz, C^1), 23.5 (d, *J* = 6.3 Hz, C^1), 23.5 (d, *J* = 4.0 Hz, C^1), 23.4 (d, *J* = 6.0 Hz, C^1), 10.0 (C^3 or C^4), 9.1 (C^3 or C^4), 7.2 (C^3 or C^4).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.92.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{30}O_5PS$: 425.1552; found: 425.1563 (Δ_{HRMS} = 3.69 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2,3-diphenylpenta-3,4-dienoate (4fa)

According to the general procedure, starting from **1f** and **3a**, product **4fa** (55 mg, 0.12 mmol, 62%) was obtained as a yellow oil.

R_f = 0.13 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.88–7.82 (m, 2 H, H^{Ar}), 7.34–7.21 (m, 5 H, H^{Ar}), 7.18–7.12 (m, 3 H, H^{Ar}), 5.34 (d, *J* = 2.6 Hz, 2 H, H^5), 4.56 (dhept, *J* = 9.0, 6.2 Hz, 1 H, H^2), 4.41 (dhept, *J* = 8.8, 6.2 Hz, 1 H, H^2), 3.65 (s, 3 H, H^6), 1.25 (d, *J* = 6.2 Hz, 3 H, H^1), 1.17 (d, *J* = 6.1 Hz, 3 H, H^1), 1.15 (d, *J* = 6.2 Hz, 3 H, H^1), 1.13 (d, *J* = 6.1 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 209.7 (C^{10}), 169.5 (d, *J* = 2.9 Hz, C^{13}), 138.5 (C^{Ar}), 133.8 (C^{Ar}), 128.9 (2 C, C^{Ar}), 127.9 (2 C, C^{Ar}), 127.3 (5 C, C^{Ar}), 126.5 (C^{Ar}), 107.6 (d, *J* = 8.7 Hz, C^{11}), 80.9 (C^5), 72.2 (d, *J* = 7.5 Hz, C^2), 72.2 (d, *J* = 6.7 Hz, C^2), 64.4 (C^{12}), 52.6 (C^6), 23.3–22.8 (m, 4 C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 18.88.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{24}H_{30}O_5PS$: 461.1552; found: 461.1564 (Δ_{HRMS} = 3.62 ppm).

Downloaded by: Fudan University Library. Copyrighted material.

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(4-fluorophenyl)-penta-3,4-dienoate (4ab)

According to the general procedure, starting from **1a** and **3b**, product **4ab** (32 mg, 0.08 mmol, 61%) was obtained as an orange oil.

R_f = 0.13 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.77–7.71 (m, 2 H, H^6), 7.30–7.21 (m, 2 H, H^7), 6.58 (t, *J* = 6.6 Hz, 1 H, H^3), 5.10 (dd, *J* = 6.7, 1.7 Hz, 2 H, H^4), 4.98 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 4.85 (dhept, *J* = 8.8, 6.2 Hz, 1 H, H^2), 4.04 (s, 3 H, H^5), 1.58 (d, *J* = 6.2 Hz, 3 H, H^1), 1.54 (d, *J* = 6.2 Hz, 6 H, H^1), 1.49 (d, *J* = 6.2 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 207.9 (C^9), 170.9 (d, *J* = 5.7 Hz, C^{11}), 162.3 (d, *J* = 248.1 Hz, C^8), 134.5 (dd, *J* = 7.2, 3.3 Hz, C^{12}), 129.9 (d, *J* = 8.3 Hz, 2 C, C^6), 115.0 (d, *J* = 21.7 Hz, 2 C, C^7), 95.9 (d, *J* = 2.6 Hz, C^3), 79.5 (C^4), 73.0 (d, *J* = 6.8 Hz, C^2), 73.0 (d, *J* = 7.5 Hz, C^2), 63.1 (d, *J* = 3.3 Hz, C^{10}), 53.5 (C^5), 23.8 (d, *J* = 3.7 Hz, C^1), 23.7 (d, *J* = 4.2 Hz, C^1), 23.5 (d, *J* = 5.6 Hz, C^1), 23.5 (d, *J* = 5.9 Hz, C^1).

^{19}F NMR (376 MHz, $CDCl_3$): δ = –113.9.

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.42.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{25}FO_5PS$: 403.1144; found: 403.1152 (Δ_{HRMS} = 2.96 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(2-fluorophenyl)-penta-3,4-dienoate (4ac)

According to the general procedure, starting from **1a** and **3c**, product **4ac** (70 mg, 0.17 mmol, 82%) was obtained as a clear oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.71 (td, *J* = 8.0, 1.7 Hz, 1 H, H^6), 7.41–7.20 (m, 1 H, H^{7-9}), 7.14 (t, *J* = 7.6 Hz, 1 H, H^{7-9}), 7.05–6.94 (m, 1 H, H^{7-9}), 6.51 (t, *J* = 6.6 Hz, 1 H, H^3), 4.85 (qd, *J* = 11.4, 6.6 Hz, 2 H, H^4), 4.74 (dhept, *J* = 8.7, 6.1 Hz, 1 H, H^2), 4.32 (dhept, *J* = 8.0, 6.1 Hz, 1 H, H^2), 3.75 (s, 3 H, H^5), 1.29 (dd, *J* = 23.9, 6.2 Hz, 6 H, H^1), 1.09 (dd, *J* = 14.8, 6.2 Hz, 6 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 206.9 (C^{12}), 170.0 (d, *J* = 9.4 Hz, C^{14}), 160.1 (d, *J* = 249.1 Hz, C^{10}), 130.1 (d, *J* = 4.3 Hz, C^7 + C^9), 130.0 (d, *J* = 1.7 Hz, C^7 + C^9), 127.2 (dd, *J* = 11.5, 3.4 Hz, C^{11}), 123.5 (d, *J* = 3.4 Hz, C^8), 115.4 (d, *J* = 21.8 Hz, C^6), 93.7 (d, *J* = 2.8 Hz, C^3), 79.3 (C^4), 73.2 (d, *J* = 6.6 Hz, C^2), 73.1 (d, *J* = 7.7 Hz, C^2), 59.7 (d, *J* = 3.3 Hz, C^{13}), 53.4 (C^5), 23.9 (d, *J* = 3.5 Hz, C^1), 23.5 (d, *J* = 5.4 Hz, 3 C, C^1).

^{19}F NMR (376 MHz, $CDCl_3$): δ = –109.7.

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.03.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{25}FO_5PS$: 403.1144; found: 403.1153 (Δ_{HRMS} = 1.39 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(4-(trifluoromethyl)-phenyl)penta-3,4-dienoate (4ad)

According to the general procedure, starting from **1a** and **3d**, product **4ad** (24 mg, 0.11 mmol, 25%) was obtained as a clear oil after reverse-phase chromatography as described above.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.65 (d, *J* = 8.4 Hz, 2 H, H^7), 7.58 (d, *J* = 8.4 Hz, 2 H, H^6), 6.36 (t, *J* = 6.6 Hz, 1 H, H^3), 4.88 (dd, *J* = 6.7, 0.9 Hz, 2 H, H^4), 4.70 (dhept, *J* = 8.8, 6.3 Hz, 1 H, H^2), 4.55 (dhept, *J* = 8.6, 6.2 Hz, 1 H, H^2), 3.79 (s, 3 H, H^5), 1.31 (d, *J* = 6.2 Hz, 3 H, H^1), 1.27 (d, *J* = 5.4 Hz, 3 H, H^1), 1.25 (d, *J* = 4.9 Hz, 3 H, H^1), 1.19 (d, *J* = 6.2 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 207.8 (C^9), 170.4 (d, *J* = 6.6 Hz, C^{12}), 142.9 (d, *J* = 6.2 Hz, C^{11}), 130.2 (q, *J* = 32.6 Hz, C^8), 128.6 (C^6), 125.0 (q, *J* = 3.8 Hz, C^7), 123.9 (q, *J* = 272.1 Hz, C^{13}), 95.2 (d, *J* = 3.0 Hz, C^3), 79.7 (C^4), 73.2 (d, *J* = 6.5 Hz, C^2), 73.1 (d, *J* = 7.1 Hz, C^2), 63.2 (d, *J* = 3.2 Hz, C^{10}), 53.6 (C^5), 23.8 (d, *J* = 3.7 Hz, C^1), 23.7 (d, *J* = 4.2 Hz, C^1), 23.5 (d, *J* = 6.0 Hz, C^1), 23.5 (d, *J* = 6.0 Hz, C^1).

^{19}F NMR (376 MHz, $CDCl_3$): δ = –62.7.

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.15.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{25}F_3O_5PS$: 453.1112; found: 453.1124 (Δ_{HRMS} = 2.65 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(3-(trifluoromethyl)-phenyl)penta-3,4-dienoate (4ae)

According to the general procedure, starting from **1a** and **3e**, product **4ae** (50 mg, 0.11 mmol, 46%) was obtained as a clear oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.80 (d, *J* = 2.0 Hz, 1 H, CH^{Ar}), 7.71 (d, *J* = 7.9 Hz, 1 H, CH^{Ar}), 7.54 (d, *J* = 7.8 Hz, 1 H, CH^{Ar}), 7.45 (t, *J* = 7.9 Hz, 1 H, CH^{Ar}), 6.33 (t, *J* = 6.6 Hz, 1 H, *CH*=C=CH_2), 4.93–4.80 (m, 2 H, CH=C=*CH*$_2$), 4.69 (dhept, *J* = 9.1, 6.2 Hz, 1 H, *CH*(CH_3)$_2$), 4.59 (dhept, *J* = 9.1, 6.2 Hz, 1 H, *CH*(CH_3)$_2$), 3.79 (s, 3 H, CO_2Me), 1.30 (d, *J* = 6.2 Hz, 3 H, CH(*CH*$_3$)$_2$), 1.26 (dd, *J* = 6.2, 1.3 Hz, 6 H, CH(*CH*$_3$)$_2$), 1.20 (d, *J* = 6.2 Hz, 3 H, CH(*CH*$_3$)$_2$).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 207.9, 170.4 (d, *J* = 6.3 Hz), 139.95 (d, *J* = 6.3 Hz), 131.5, 130.4 (q, *J* = 32.4 Hz), 128.5, 125.2 (q, *J* = 3.9 Hz), 124.8 (q, *J* = 3.7 Hz), 123.9 (q, *J* = 272.4 Hz), 95.4 (d, *J* = 3.2 Hz), 79.7, 73.2 (d, *J* = 6.5 Hz), 73.1 (d, *J* = 7.1 Hz), 63.2 (d, *J* = 3.3 Hz), 53.5, 23.7 (d, *J* = 3.8 Hz), 23.6 (d, *J* = 4.2 Hz), 23.5 (d, *J* = 2.9 Hz), 23.4 (d, *J* = 3.2 Hz).

^{19}F NMR (376 MHz, $CDCl_3$): δ = –62.6.

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.02.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{25}F_3O_5PS$: 453.1112; found: 453.1119 (Δ_{HRMS} = 1.54 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(4-nitrophenyl)penta-3,4-dienoate (4af)

According to the general procedure, starting from **1a** and **3f**, product **4af** (40 mg, 0.09 mmol, 44%) was obtained as a yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.22–8.07 (m, 2 H, H^7), 7.77–7.63 (m, 2 H, H^6), 6.32 (t, *J* = 6.6 Hz, 1 H, H^3), 4.97–4.83 (m, 2 H, H^4), 4.69 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 4.58 (dhept, *J* = 8.7, 6.2 Hz, 1 H, H^2), 3.79 (s, 3 H, H^5), 1.30 (d, *J* = 6.2 Hz, 3 H, H^1), 1.26 (d, *J* = 6.2 Hz, 6 H, H^1), 1.20 (d, *J* = 6.2 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 207.8 (C^9), 170.0 (d, *J* = 6.7 Hz, C^{12}), 147.3 (C^8), 146.0 (d, *J* = 5.6 Hz, C^{11}), 129.4 (C^6 or C^7), 123.1 (C^6 or C^7), 95.0 (d, *J* = 3.5 Hz, C^3), 80.0 (C^4), 73.4 (d, *J* = 6.3 Hz, C^2), 73.3 (d, *J* = 6.7 Hz, C^2), 63.0 (d, *J* = 3.2 Hz, C^{10}), 53.7 (C^5), 23.8 (d, *J* = 3.8 Hz, C^1), 23.7 (d, *J* = 4.2 Hz, C^1), 23.5 (d, *J* = 4.0 Hz, C^1), 23.5 (d, *J* = 4.4 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 18.52.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{25}NO_7PS$: 430.1089; found: 430.1095 (Δ_{HRMS} = 1.39 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-(3-methoxyphenyl)-penta-3,4-dienoate (4ag)

According to the general procedure, starting from **1a** and **3g**, product **4ag** (41 mg, 0.10 mmol, 48%) was obtained as a pale-yellow oil.

R_f = 0.10 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.27–7.20 (m, 1 H, H^{Ar}), 7.09–7.04 (m, 2 H, H^{Ar}), 6.84–6.79 (m, 1 H, H^{Ar}), 6.34 (t, *J* = 6.6 Hz, 1 H, H^3), 4.89–4.81 (m, 2 H, H^4), 4.76 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 4.61 (dhept, *J* = 8.8, 6.2 Hz, 1 H, H^2), 3.79 (s, 3 H, H^5 or H^6), 3.78 (s, 3 H, H^5 or H^6), 1.33 (d, *J* = 6.2 Hz, 3 H, H^1), 1.29 (dd, *J* = 6.2, 2.4 Hz, 6 H, H^1), 1.24 (d, *J* = 6.2 Hz, 3 H, H^1).

© 2025. Thieme. All rights reserved.

Downloaded by: Fudan University Library. Copyrighted material.

^{13}C NMR (101 MHz, $CDCl_3$): δ = 208.0 (C^7), 170.9 (d, *J* = 5.5 Hz, C^9), 159.3 (C^{Ar}), 140.0 (d, *J* = 7.8 Hz, C^{Ar}), 129.1 (C^{Ar}), 120.1 (d, *J* = 1.0 Hz, C^{Ar}), 114.0 (d, *J* = 1.0 Hz, C^{Ar}), 113.5 (C^{Ar}), 95.9 (d, *J* = 2.1 Hz, C^3), 79.3 (C^4), 73.0 (d, *J* = 6.5 Hz, C^2), 72.9 (d, *J* = 7.1 Hz, C^2), 63.5 (d, *J* = 3.3 Hz, C^8), 55.2 (C^5 or C^6), 53.4 (C^5 or C^6), 23.9 (d, *J* = 3.7 Hz, C^1), 23.7 (d, *J* = 4.2 Hz, C^1), 23.6 (d, *J* = 5.5 Hz, C^1), 23.5 (d, *J* = 5.7 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.66.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{28}O_6PS$: 415.1344; found: 415.1353 (Δ_{HRMS} = 3.29 ppm).

(1*R*,2*S*,5*R*)-2-Isopropyl-5-methylcyclohexyl 2-((Diisopropoxyphosphoryl)thio)-2-phenylpenta-3,4-dienoate (4ai)

According to the general procedure, starting from **1a** and **3i**, product **4ai** (60 mg, 0.12 mmol, 56%) was obtained as a yellow oil after silica gel chromatography (10–20% Et_2O in pentane) as a mixture of non-separable diastereoisomers (dr 66:34).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.55–7.45 (m, 2 H, 2 H^5), 7.31–7.15 (m, 3 H, 2 H^6 + H^7), 6.37 (t, *J* = 6.7 Hz, H^3-*major*), 6.32 (t, *J* = 6.6 Hz, H^3-*minor*), 4.86 (m, 2 H^4-*major*), 4.78 (m, 2 H^4-*minor*), 4.75–4.31 (m, 3 H, H^8 + 2 H^2), 2.10–1.99 (m, 1 H), 1.62–1.52 (m, 2 H), 1.47–1.31 (m, 2 H), 1.29–1.14 (d, *J* = 6.1 Hz, 12 H), 1.08 (d, *J* = 6.2 Hz, 3 H^1-*minor*), 0.99–0.67 (m, 8 H), 0.62–0.56 (m, 3 H), 0.50 (d, *J* = 6.8 Hz, 3 H, H^{16}-*minor*).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 208.2 (C^{17}-*minor*), 207.7 (C^{17}-*major*), 169.9 (d, *J* = 6.2 Hz, C^{20}-*minor*), 169.7 (d, *J* = 8.8 Hz, C^{20}-*major*), 139.6 (d, *J* = 4.9 Hz, C^{18}-*major*), 138.9 (d, *J* = 6.6 Hz, C^{18}-*minor*), 128.1–127.7 (m, 5 C, C^5 + C^6 + C^7), 95.8 (C^3-*minor*), 94.6 (C^3-*major*), 79.4 (C^4-*major*), 79.3 (C^4-*minor*), 77.3 (d, *J* = 3.9 Hz, C^8-*major*), 77.0 (C^8-*minor*), 72.8–72.4 (m, 2 C^2), 64.2 (d, *J* = 2.8 Hz, C^{19}-*major*), 63.4 (d, *J* = 3.3 Hz, C^{19}-*minor*), 46.94 (*minor*), 46.90 (*major*), 40.3 (*major*), 40.1 (*minor*), 34.17 (*minor*), 34.15 (*major*), 31.41 (*major*), 31.38 (*minor*), 25.4 (*minor*), 25.2 (*major*), 23.9 (d, *J* = 3.6 Hz, 1 C^1-*minor*), 23.9 (d, *J* = 3.6 Hz, 1 C^1-*major*), 23.7 (d, *J* = 3.8 Hz, C^1), 23.6–23.5 (m, 2 C^1), 23.5 (1 C^1-*major*), 23.4 (1 C^1-*minor*), 22.9, 22.0, 20.8 (*minor*), 20.6 (*major*), 15.74 (*minor*), 15.72 (*major*).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.06 (*major*), 19.78 (*minor*).

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{42}O_5PS$: 509.2491; found: 509.2496 (Δ_{HRMS} = 0.98 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2-phenylpent-4-enoate (5aa)

According to the general procedure, starting from **2a** and **3a**, product **5aa** (67 mg, 0.17 mmol, 83%) was obtained as a pale-yellow oil.

R_f = 0.22 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.47–7.43 (m, 2 H, H^7), 7.36–7.30 (m, 2 H, H^8), 7.29–7.23 (m, 1 H, H^9), 5.87 (ddt, *J* = 17.1, 10.2, 6.9 Hz, 1 H, H^4), 5.16–5.07 (m, 2 H, H^5), 4.71 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 4.35 (dhept, *J* = 8.5, 6.2 Hz, 1 H, H^2), 3.73 (s, 3 H, H^6), 3.72–3.66 (m, 1 H, H^3), 3.20 (dd, *J* = 14.4, 7.4 Hz, 1 H, H^3), 1.33 (d, *J* = 6.2 Hz, 3 H, H^1), 1.29 (d, *J* = 6.2 Hz, 3 H, H^1), 1.13 (d, *J* = 6.3 Hz, 3 H, H^1), 1.11 (d, *J* = 6.3 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 172.0 (d, *J* = 9.0 Hz, C^{11}), 139.6 (d, *J* = 6.1 Hz, C^{12}), 133.4 (C^4), 128.3 (2 C, C^7 or C^8), 127.9 (C^9), 127.3 (2 C, C^7 or C^8), 119.0 (C^5), 72.8 (d, *J* = 6.5 Hz, C^2), 72.7 (d, *J* = 7.5 Hz, C^2), 64.7 (d, *J* = 2.4 Hz, C^{10}), 53.1 (C^6), 41.8 (d, *J* = 1.2 Hz, C^3), 23.9 (d, *J* = 3.7 Hz, C^1), 23.6 (d, *J* = 4.5 Hz, C^1), 23.5 (d, *J* = 4.1 Hz, C^1), 23.5 (d, *J* = 3.7 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.02.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{28}O_5PS$: 387.1395; found: 387.1399 (Δ_{HRMS} = 2.18 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-3-methyl-2-phenylpent-4-enoate (5ba)

According to the general procedure, starting from **2b** and **3a**, product **5ba** (103 mg, 0.26 mmol, 65%) was obtained as a colorless oil as a mixture of nonseparable diastereoisomers (dr 53:47).

R_f = 0.20 (*n*-pentane/EtOAc, 6:4).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.45 (t, *J* = 7.3 Hz, 2 H, H^{10}), 7.34–7.17 (m, 3 H, H^{11} + H^{12}), 5.72 (dddd, *J* = 20.3, 17.8, 10.3, 7.9 Hz, 1 H, H^4), 5.07 (dd, *J* = 23.7, 10.1 Hz, 2 H, H^5), 4.56 (dddt, *J* = 30.0, 12.3, 8.7, 6.2 Hz, 2 H, H^2), 3.83 (s, H^8), 3.81 (s, H^8), 3.61 (q, *J* = 6.4 Hz, 1 H, H^3), 1.27–1.12 (m, 12 H, H^1), 1.09 (d, *J* = 6.8 Hz, H^6), 1.05 (d, *J* = 6.7 Hz, H^6).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 171.0 (d, *J* = 5.1 Hz, C^{13}-*minor*), 170.8 (d, *J* = 4.9 Hz, C^{13}-*major*), 138.9 (C^4-*major*), 138.7 (C^4-*minor*), 137.5 (d, *J* = 5.2 Hz, C^9-*major*), 137.2 (d, *J* = 5.1 Hz, C^9-*minor*), 129.2 (2 CH^{Ar}-*minor*), 129.1 (2 CH^{Ar}-*major*), 127.5 (C^{12}-*minor*), 127.5 (C^{12}-*major*), 127.4 (2 CH^{Ar}-*major*), 127.3 (2 CH^{Ar}-*minor*), 116.9 (C^5-*major*), 116.8 (C^5-*minor*), 73.1–72.8 (m, 2 C^2), 67.9 (d, *J* = 4.1 Hz, C^7-*major*), 67.6 (d, *J* = 4.0 Hz, C^7-*minor*), 52.8 (C^8-*minor*), 52.6 (C^8-*major*), 45.2 (d, *J* = 5.6 Hz, C^3-*minor*), 45.1 (d, *J* = 5.8 Hz, C^3-*major*), 23.9 (t, *J* = 3.8 Hz, 2 C^1), 23.7 (dd, *J* = 4.2, 1.9 Hz, 2 C^1), 23.6 (d, *J* = 5.7 Hz, 2 C^1), 23.4 (d, *J* = 5.0 Hz, C^1), 23.3 (d, *J* = 4.8 Hz, C^1), 17.3 (C^6-*minor*), 16.4 (C^6-*major*).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.66 (*major*), 19.63 (*minor*).

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{30}O_5PS$: 401.1546; found: 401.1560 (Δ_{HRMS} = –3.00 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-2,3-diphenylpent-4-enoate (5ca)

According to the general procedure, starting from **2c** and **3a**, product **5ca** (37 mg, 0.08 mmol, 50%) was obtained as a colorless oil as a mixture of nonseparable diastereoisomers (dr 54:46) after reverse-phase chromatography as described above.

R_f = 0.20 (*n*-pentane/Et_2O, 1:1).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.40–7.33 (m, 2 H, H^{Ar}-*major*), 7.33–7.21 (m, 4 H, H^{Ar}), 7.13 (dt, *J* = 21.2, 6.6 Hz, 3 H, H^{Ar}), 6.93 (dd, *J* = 7.6, 1.8 Hz, 2 H, H^{Ar}-*minor*), 6.71 (d, *J* = 7.2 Hz, 2 H, H^{Ar}-*major*), 6.22 (dt, *J* = 17.9, 8.7 Hz, H^4-*minor*), 5.99 (dt, *J* = 16.9, 10.0 Hz, H^4-*major*), 5.26–5.12 (m, 2 H, H^5), 4.90 (d, *J* = 8.6 Hz, H^3-*minor*), 4.69 (ddt, *J* = 12.4, 8.4, 6.2 Hz, H^2-*minor*), 4.62 (d, *J* = 9.3 Hz, H^3-*major*), 4.55 (ddd, *J* = 11.1, 8.4, 5.7 Hz, H^2-*major*), 3.86 (s, H^8-*major*), 3.81 (s, H^8-*minor*), 1.28–1.14 (m, 9 H, H^1), 1.10 (d, *J* = 6.2 Hz, 3 H, H^1-*minor*), 1.03 (d, *J* = 6.2 Hz, 3 H, H^1-*major*).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 170.9 (d, *J* = 5.7 Hz, C^{13}-*minor*), 170.8 (d, *J* = 3.6 Hz, C^{13}-*major*), 139.7 (C^6-*major*), 137.9 (C^4-*minor*), 137.4 (d, *J* = 4.3 Hz, C^9-*minor*), 137.0 (C^4-*major*, C^6-*minor*), 135.3 (d, *J* = 3.8 Hz, C^9-*major*), 131.0 (2 CH^{Ar}-*major*), 130.8 (2 CH^{Ar}-*major*), 130.5 (2 CH^{Ar}-*minor*), 129.9 (2 CH^{Ar}-*minor*), 127.8 (CH^{Ar}-*major*), 127.7 (CH^{Ar}-*minor*), 127.6 (2 CH^{Ar}-*minor*), 127.5 (2 CH^{Ar}-*major*), 127.4 (CH^{Ar}-*major*), 127.3 (2 CH^{Ar}-*minor*), 127.2 (CH^{Ar}-*minor*), 126.8 (2 CH^{Ar}-*major*), 118.9 (C^5-*major*), 118.7 (C^5-*minor*), 73.6 (d, *J* = 7.6 Hz, C^2-*major*), 73.5 (d, *J* = 7.3 Hz, C^2-*major*), 73.1 (d, *J* = 6.8 Hz, C^2-*minor*), 73.1 (d, *J* = 6.8 Hz, C^2-*minor*), 67.9 (d, *J* = 3.9 Hz, C^7-*minor*), 67.1 (d, *J* = 4.7 Hz, C^7-*major*), 58.2 (d, *J* = 8.9 Hz, C^3-*major*), 56.6 (d, *J* = 5.8 Hz, C^3-*minor*), 53.0 (C^8-*major*), 52.8 (C^8-*minor*), 24.0 (dd, *J* = 3.7, 1.4 Hz, 2 C^1), 23.9 (d, *J* = 4.2 Hz, C^1), 23.8–23.6 (m, 3 C^1), 23.4 (d, *J* = 6.2 Hz, C^1-*minor*), 23.3 (d, *J* = 6.2 Hz, C^1-*major*).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.40 (*minor*), 19.27 (*major*).

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{24}H_{32}O_5PS$: 463.1703; found: 463.1721 (Δ_{HRMS} = 3.77 ppm).

Downloaded by: Fudan University Library. Copyrighted material.

Methyl 2-((Diisopropoxyphosphoryl)thio)-3,3-dimethyl-2-phenylpent-4-enoate (5da)

According to the general procedure, starting from **2d** and **3a**, product **5da** (64 mg, 0.15 mmol, 82%) was obtained as a colorless oil.

R_f = 0.20 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.64–7.57 (m, 2 H, H^{Ar}), 7.31–7.19 (m, 3 H, H^{Ar}), 6.21 (dd, *J* = 17.4, 10.8 Hz, 1 H, H^4), 5.02 (d, *J* = 10.8 Hz, 1 H, H^5), 4.89 (d, *J* = 17.4 Hz, 1 H, H^5), 4.72 (ddt, *J* = 12.4, 8.7, 6.2 Hz, 1 H, H^2), 4.46 (ddt, *J* = 12.4, 8.7, 6.2 Hz, 1 H, H^2), 3.76 (s, 3 H, H^8), 1.27 (dd, *J* = 6.2, 2.7 Hz, 6 H, H^1), 1.22 (s, 3 H, H^6), 1.21–1.17 (m, 6 H, H^6 + H^1), 1.06 (d, *J* = 6.2 Hz, 3 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 170.4 (d, *J* = 3.3 Hz, C^{13}), 143.9 (C^4), 138.0 (d, *J* = 3.4 Hz, C^9), 130.4 (2 C^{Ar}), 127.3 (C^{12}), 126.5 (2 C^{Ar}), 113.9 (C^5), 72.9 (d, *J* = 7.3 Hz, C^2), 72.8 (d, *J* = 7.2 Hz, C^2), 71.6 (d, *J* = 5.9 Hz, C^7), 52.3 (C^8), 46.0 (d, *J* = 7.2 Hz, C^3), 24.8 (d, *J* = 4.8 Hz, C^1), 23.9–23.8 (m, C^1), 23.7 (d, *J* = 5.7 Hz, C^6), 23.2 (d, *J* = 6.4 Hz, C^6).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.47.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{20}H_{32}O_5PS$: 415.1703; found: 415.1720 (Δ_{HRMS} = 3.81 ppm).

Methyl 2-((Diisopropoxyphosphoryl)thio)-4-methyl-2-phenylpent-4-enoate (5ea)

According to the general procedure, starting from **2e** and **3a**, product **5ea** (35 mg, 0.09 mmol, 44%) was obtained as a colorless oil after a second purification using reverse-phase chromatography as described above.

R_f = 0.24 (*n*-pentane/EtOAc, 6:4).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.31–7.23 (m, 2 H, H^6 or H^7), 7.14–7.05 (m, 2 H, H^6 or H^7), 7.07–6.98 (m, 1 H, H^8), 4.68 (t, *J* = 1.8 Hz, 1 H, H^4), 4.66–4.60 (m, 1 H, H^4), 4.44 (dhept, *J* = 9.0, 6.2 Hz, 1 H, H^2), 4.06 (dhept, *J* = 8.2, 6.2 Hz, 1 H, H^2), 3.51 (d, *J* = 14.8 Hz, 1 H, H^3), 3.50 (s, 3 H, H^9), 2.98 (d, *J* = 14.4 Hz, 1 H, H^3), 1.45 (s, 3 H, H^5), 1.05 (dd, *J* = 15.9, 6.2 Hz, 6 H, H^1), 0.86 (dd, *J* = 17.2, 6.2 Hz, 6 H, H^1).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 172.2 (d, *J* = 9.9 Hz, C^{10}), 141.0 (C^{13}), 140.5 (d, *J* = 5.0 Hz, C^{11}), 128.1 (C^6 or C^7), 127.7 (C^8), 127.4 (C^6 or C^7), 116.3 (C^4), 72.7 (d, *J* = 3.5 Hz, C^2), 72.6 (d, *J* = 4.4 Hz, C^2), 64.1 (d, *J* = 2.5 Hz, C^{12}), 53.0 (C^9), 44.4 (d, *J* = 1.9 Hz, C^3), 24.0 (C^5), 23.9 (d, *J* = 3.6 Hz, C^1), 23.6 (d, *J* = 4.4 Hz, C^1), 23.5 (d, *J* = 4.2 Hz, C^1), 23.5 (d, *J* = 3.8 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 20.12.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{30}O_5PS$: 401.1546; found: 401.1555 (Δ_{HRMS} = 2.24 ppm).

Methyl 4-Bromo-2-((diisopropoxyphosphoryl)thio)-2-phenylpent-4-enoate (5fa)

According to the general procedure, starting from **2f** and **3a**, product **5fa** (74 mg, 0.15 mmol, 67%) was obtained as a colorless oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.30–7.24 (m, 2 H, H^6 or H^7), 7.17–7.09 (m, 2 H, H^6 or H^7), 7.09–7.02 (m, 1 H), 5.73 (q, *J* = 1.2 Hz, 1 H), 5.46 (d, *J* = 1.6 Hz, 1 H), 4.49 (dhept, *J* = 9.1, 6.2 Hz, 1 H), 4.10 (d, *J* = 15.3 Hz, 1 H), 4.00 (dhept, *J* = 8.2, 6.2 Hz, 1 H), 3.52 (s, 3 H), 3.41 (d, *J* = 15.2 Hz, 1 H), 1.10 (d, *J* = 6.2 Hz, 3 H), 1.07 (d, *J* = 6.2 Hz, 3 H), 0.86 (d, *J* = 6.1 Hz, 3 H), 0.79 (d, *J* = 6.2 Hz, 3 H).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 171.5 (d, *J* = 11.0 Hz), 139.3 (d, *J* = 5.0 Hz), 128.4, 128.1, 127.5, 127.2, 123.0, 72.9 (d, *J* = 6.6 Hz), 72.8 (d, *J* = 7.7 Hz), 63.4 (d, *J* = 2.3 Hz), 53.4, 46.6, 23.9 (d, *J* = 3.6 Hz), 23.6 (d, *J* = 4.2 Hz), 23.5 (d, *J* = 2.6 Hz), 23.4 (d, *J* = 5.9 Hz).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 19.79.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{27}BrO_5PS$: 465.0500; found: 465.0512 (Δ_{HRMS} = 2.58 ppm); isotopic Br-81 mass: 467.0494.

***S*-(5,5-Dimethyl-2-oxo-3-phenyltetrahydrofuran-3-yl) *O*,*O*-Diisopropyl Phosphorothioate (6ea)**

To a solution of substrate **5ea** (30 mg, 0.075 mmol, 1 equiv) in CH_2Cl_2 (0.1 M) was added *p*-toluenesulfonic acid monohydrate (13 mg, 0.075 mmol, 1 equiv). The reaction mixture was then stirred at 60 °C in a sealed vial until completion, and the solvent was removed under vacuum. The crude was purified by flash chromatography on silica gel (*n*-pentane/Et_2O) to get pure product **6ea** (18 mg, 0.047 mmol, 62%) as a colorless oil.

R_f = 0.25 (*n*-pentane/Et_2O, 7:3).

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.71–7.65 (m, 2 H, H^5 or H^6), 7.41–7.35 (m, 2 H, H^5 or H^6), 7.34–7.29 (m, 1 H, H^7), 4.92 (dhept, *J* = 9.3, 6.2 Hz, 1 H, H^2), 4.54 (dhept, *J* = 9.1, 6.2 Hz, 1 H, H^2), 3.34 (d, *J* = 13.5 Hz, 1 H, H^4), 3.17 (d, *J* = 13.5 Hz, 1 H, H^4), 1.58 (s, 3 H, H^3), 1.39 (d, *J* = 6.2 Hz, 3 H, H^1), 1.33 (dd, *J* = 6.2, 0.7 Hz, 3 H, H^1), 1.27 (d, *J* = 6.2 Hz, 3 H, H^1), 1.23 (d, *J* = 6.2 Hz, 3 H, H^1), 1.05 (s, 3 H, H^3).

^{13}C NMR (101 MHz, $CDCl_3$): δ = 173.9 (d, *J* = 5.1 Hz, C^{10}), 138.6 (d, *J* = 9.5 Hz, C^{11}), 129.0 (C^5), 128.6 (C^7), 127.2 (C^6), 82.5 (C^9), 73.5 (d, *J* = 6.3 Hz, C^2), 73.1 (d, *J* = 7.3 Hz, C^2), 60.3 (d, *J* = 3.7 Hz, C^8), 49.6 (C^4), 29.1 (C^3), 27.9 (C^3), 24.0 (d, *J* = 3.6 Hz, C^1), 23.6 (d, *J* = 4.3 Hz, C^1), 23.5 (d, *J* = 3.2 Hz, C^1), 23.5 (d, *J* = 6.2 Hz, C^1).

^{31}P NMR (162 MHz, $CDCl_3$): δ = 18.15.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{18}H_{28}O_5PS$: 387.1395; found: 387.1400 (Δ_{HRMS} = 2.49 ppm).

Conflict of Interest

The authors declare no conflict of interest.

Funding Information

The authors acknowledge the financial support from the Ministère de l'Enseignement Supérieur et de la Recherche (doctoral grant to L.T. and C.R.), the Centre National de la Recherche Scientifique (CNRS) and the University of Strasbourg. This work has been published under the framework of the IdEx Unistra and benefits from a funding from the state managed by the Agence Nationale de la Recherche (French National Research Agency) as part of the Investments for the Future Program (postdoctoral grant to T.M.).

Acknowledgment

We thank the platform PACSI (GDS 3670) for NMR and mass analyses, and Pierre Locquet for helpful preliminary experiments.

Supporting Information

Supporting information for this article is available online at https://doi.org/10.1055/a-2515-0351.

Downloaded by: Fudan University Library. Copyrighted material.

References

(1) (a) Doyle, M. P. *Chem. Rev.* **1986**, *86*, 919. (b) Doyle, M. P.; Forbes, D. C. *Chem. Rev.* **1998**, *98*, 911. (c) *Sulfur-Mediated Rearrangements II*, In *Topics in Current Chemistry, Vol. 275*; Schaumann, E., Ed.; Springer: Berlin/Heidelberg/New York, **2007**. (d) Neuhaus, J. D.; Oost, R.; Merad, J.; Maulide, N. *Top. Curr. Chem.* **2018**, *376*, 15. (e) West, T. H.; Spoehrle, S. S. M.; Kasten, K.; Taylor, J. E.; Smith, A. D. *ACS Catal.* **2015**, *5*, 7446. (f) Jana, S.; Guo, Y.; Koenigs, R. M. *Chem. Eur. J.* **2021**, *27*, 1270. (g) Shi, C.-Y.; Zhou, B.; Teng, M.-Y.; Ye, L.-W. *Synthesis* **2023**, *55*, 3895. (h) Hommelsheim, R.; Guo, Y.; Yang, Z.; Empel, C.; Koenigs, R. M. *Angew. Chem. Int. Ed.* **2019**, *58*, 1203. (i) Orłowska, K.; Rybicka-Jasińska, K.; Krajewski, P.; Gryko, D. *Org. Lett.* **2020**, *22*, 1018.

(2) Zhang, Z.; Sheng, Z.; Yu, W.; Wu, G.; Zhang, R.; Chu, W.-D.; Zhang, Y.; Wang, J. *Nat. Chem.* **2017**, *9*, 970.

(3) Rognan, C.; Locquet, P.; Pertschi, R.; Girard, N.; Gulea, M. *Adv. Synth. Catal.* **2024**, *366*, 4078.

(4) Liu, X.-S.; Tang, Z.; Li, Z.; Li, M.; Xu, L.; Liu, L. *Nat. Commun.* **2021**, *12*, 7298.

(5) (a) Ung, S. P.-M.; Li, C.-J. *RSC Sustainability* **2023**, *1*, 11. (b) Quin, L. D. *A Guide to Organophosphorus Chemistry*; Wiley Interscience: New York, **2000**.

(6) See, for examples: (a) Eckstein, F. *Antisense Nucleic Acid Drug Dev.* **2000**, *10*, 117. (b) Zhang, S.; Zhang, M.-H.; Feng, S.; Zhang, W.-J.; Zhu, Y.-Y.; Li, Z.-W.; Bai, S. *Chem. Pap.* **2024**, *78*, 4045.

(7) (a) Jones, D. J.; O'Leary, E. M.; O'Sullivan, T. P. *Tetrahedron Lett.* **2018**, *59*, 4279. (b) Gulea, M. *Adv. Org. Synth.* **2018**, *12*, 117.

(8) (a) Jones, D. J.; O'Leary, E. M.; O'Sullivan, T. P. *Adv. Synth. Catal.* **2020**, *362*, 2801. (b) Zheng, H.; Peng, J.; Liu, X.; Zhao, P. *Adv. Synth. Catal.* **2024**, *366*, 3739.

(9) Huang, M.; Ye, Z.; Xu, W.; Xu, M.-H. *Asian J. Org. Chem.* **2023**, *12*, e202300053.

(10) For anionic S-to-C phosphoryl group migration, see: (a) Masson, S.; Saquet, M.; Marchand, P. *Tetrahedron* **1998**, *54*, 1523. (b) Philippitsch, V.; Hammerschmidt, F. *Org. Biomol. Chem.* **2011**, *9*, 5220.

(11) (a) Mattiza, J. T.; Fohrer, J. G. G.; Duddeck, H.; Gardiner, M. G.; Ghanem, A. *Org. Biomol. Chem.* **2011**, *9*, 6542. (b) Zhu, D.; Ma, J.; Luo, K.; Fu, H.; Zhang, L.; Zhu, S. *Angew. Chem. Int. Ed.* **2016**, *55*, 8452.

(12) Davies, P. W.; Albrecht, S. J.-C.; Assanelli, G. *Org. Biomol. Chem.* **2009**, *7*, 1276.

Downloaded by: Fudan University Library. Copyrighted material.